

Arjun Giriraj Day 12

1. What is a network interface?
Network interface lets the system communicate with the network. It can have different names on different systems.
2. What is the loopback interface?
A loopback interface allows system to talk to itself, it has IPv4 address of 127.0.0.1.
3. What does ip addr display?
Ip addr displays the interface name along with IP addresses.
4. What is the purpose of ip link?
ip link displays the interface with link layer info as state, mac address, and mtu.
5. What does hostname -I display?
It displays the system's ip address.
6. What does ip route show?
It shows the network route, default IP address via gateway with which interface used.
7. What is a default route?
Default route is the route network use when the ip route table does not have data of the destination address so the default route comes in use.
8. What does ping test?
Ping checks the connectivity and checks if the destination is accessible.
9. Why can a failed ping be misleading?
Failed ping can be mislead because ICMP is blocked by firewalls.
10. What is DNS?
Domain name server, is is like a record of each domain name with its IP address.
11. What does getent hosts do?
It shows the ip address with hostname.
12. What information does ip neigh display?
It shows the neighbouring system which contains mapping between ip address and mac address for host.
13. What is a socket?
It is the endpoint for the computer networking, it serves as the interface for receiving and sending data.
14. What does ss -tuln display?
Shows listening tcp/upd sockets.

15. What is the difference between TCP and UDP conceptually?
TCP is the reliable, and secured protocol; While UDP is not reliable and secured but generally faster than TCP.
16. What is curl used for?
Curl command used to fetch the page content.
17. What does curl -I request?
It used to fetch only http response header.
18. What is traceroute used for?
It shows the hop route towards the destination.
19. Why is ss commonly preferred over netstat on modern Linux?
Its because ss is generally faster, more feature and pre installed on linux systems.
20. Why are Linux networking commands useful in cybersecurity?
Professional use these commands for threat hunting, soc investigation, incident response and more.

TROUBLESHOOTING QUESTION (from lab Task 10):

1. A Linux system can ping 8.8.8.8, but ping google.com fails because the hostname cannot be resolved. What type of problem is most likely, and which command from today would you use first to confirm it?
It could be the dns resolution failure, and to confirm that we must run the command nslookup [google.com](https://www.google.com) and if it return no answer then the issue is confirmed.

PRACTICAL SCREENSHOTS (from your own VM/lab only):

ip addr · ip route · ping -c 4 127.0.0.1 · getent hosts google.com · ss -tuln · curl -I <https://example.com>

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venom@VENOM:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: wlan0: <BROADCAST,MULTICAST,UP,LOWER UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 10:6f:d9:7a:da:a9 brd ff:ff:ff:ff:ff:ff
    inet 10.170.133.85/24 brd 10.170.133.255 scope global dynamic noprefixroute wlan0
        valid_lft 2762sec preferred_lft 2762sec
    inet6 2401:4900:bb69:8bdc:6f08:297:a405:effe/64 scope global dynamic noprefixroute
        valid_lft 7024sec preferred_lft 7024sec
    inet6 fe80::83a4:13e:661d:1ec/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
venom@VENOM:~$ ip route
default via 10.170.133.116 dev wlan0 proto dhcp src 10.170.133.85 metric 600
10.170.133.0/24 dev wlan0 proto kernel scope link src 10.170.133.85 metric 600
venom@VENOM:~$ ping -c 4 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.043 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.047 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.051 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.047 ms

--- 127.0.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3052ms
rtt min/avg/max/mdev = 0.043/0.047/0.051/0.003 ms
venom@VENOM:~$ getent hosts google.com
2404:6800:4013:813::71 google.com
2404:6800:4013:813::8a google.com
2404:6800:4013:813::65 google.com
2404:6800:4013:813::66 google.com

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venom@VENOM:~$ ss -tuln
Netid  State  Recv-Q  Send-Q  Local Address:Port  Peer Address:Port
udp    UNCONN  0        0        224.0.0.251:5353    0.0.0.0:*
udp    UNCONN  0        0        224.0.0.251:5353    0.0.0.0:*
udp    UNCONN  0        0        224.0.0.251:5353    0.0.0.0:*
udp    UNCONN  0        0        0.0.0.0:5355        0.0.0.0:*
udp    UNCONN  0        0        0.0.0.0:55104       0.0.0.0:*
udp    UNCONN  0        0        127.0.0.54:53       0.0.0.0:*
udp    UNCONN  0        0        127.0.0.53%lo:53    0.0.0.0:*
udp    UNCONN  0        0        [::]:5355           [::]:*
udp    UNCONN  0        0        *:49067              *:*
udp    UNCONN  0        0        *:1716               *:*
tcp    LISTEN  0        4096        0.0.0.0:5355        0.0.0.0:*
tcp    LISTEN  0        4096        127.0.0.1:11434     0.0.0.0:*
tcp    LISTEN  0        4096        127.0.0.53%lo:53    0.0.0.0:*
tcp    LISTEN  0        4096        127.0.0.54:53       0.0.0.0:*
tcp    LISTEN  0        4096        [::]:5355           [::]:*
tcp    LISTEN  0        50         *:1716               *:*

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venom@VENOM:~$ curl -l google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html; charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="http://www.google.com/">here</A>.
</BODY></HTML>

```